

ZOMATO ANALYTICS PROJECT

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HELLO EVERYONE

As an aspiring data analyst, I applied my SQL and Power BI skills to analyze Zomato's restaurant, menu, and order data, uncovering key insights into cuisine performance, pricing trends, customer ordering behavior, and restaurant contributions.

This Project transforms raw data into business-ready insights that highlight which cuisines, menu items, and cities drive the most revenue, helping Zomato make data-driven decisions to optimize offerings and improve customer engagement.





BUSINESS PROBLEM



The business problem centers on understanding how restaurants, menus, and customers influence overall sales and profitability, with the goal of helping a food delivery platform make smarter decisions. The company needs to know which cuisines and menu items bring in the most revenue and profit, how pricing affects customer demand, and whether customers tend to be loyal or mostly one-time users.

It's also important to assess performance at both the restaurant and city levels to identify top performers and regional growth opportunities. Spotting anomalies, like invalid orders, ensures more accurate reporting and analysis. By combining SQL-driven insights with an interactive dashboard, this project highlights key profit drivers, customer retention patterns, and market opportunities that can inform strategies around pricing, marketing, and expansion.

This project helped me sharpen my SQL skills by working with real-world data and uncovering patterns in customer behavior, restaurant performance, and sales. I also learned how to design a clear, interactive dashboard in Power BI and, most importantly, how to turn raw data into insights that can actually guide business decisions in the food delivery space.

```
[1]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
[2]: df = pd.read_excel(
    r'users.xlsx' )
```

```
[3]: df.head()
```

```
[3]:
```

	user_id	name	Age	Gender	Marital Status	Occupation
0	1	Claire Ferguson	20	Female	Single	Student
1	2	Jennifer Young	24	Female	Single	Student
2	3	Jermaine Roberson	22	Male	Single	Student
3	4	Rachel Carpenter	22	Female	Single	Student
4	5	Shawn Parker	22	Male	Single	Student

```
[4]: df1 = pd.read_csv(r'encoded-food.csv')
```

```
[5]: df1.head()
```

```
[5]:
```

	f_id	item	veg_or_non_veg
0	fd0	Aloo Tikki Burger	Veg
1	fd1	Veg Creamy Burger	Veg
2	fd2	Cheese Burst Burger	Veg
3	fd3	Paneer Creamy Burger	Veg
4	fd4	Maxican Burger	Veg

DATA EXPLORATION

```
[6]: df2 = pd.read_csv(r"C:\Users\ASUS\OneDrive\Desktop\encoded-menu (1).csv")
```

```
[7]: df2.head()
```

```
[7]:
```

	menu_id	r_id	f_id	cuisine	price
0	mn0	567335	fd0	Beverages,Pizzas	40.0
1	mn0	567335	fd669322	Beverages,Pizzas	40.0
2	mn328	158203	fd0	Beverages	65.0
3	mn328	158203	fd669322	Beverages	65.0
4	mn449	158203	fd0	Beverages	65.0




```
[8]: df3 = pd.read_excel(r"C:\Users\ASUS\Downloads\orders.xlsx")
```

```
[9]: df3.head()
```

```
[9]:
```

	order_date	sales_qty	sales_amount	currency	user_id	r_id
0	2017-10-10	100	41241	INR	49226	567335.0
1	2018-05-08	3	-1	INR	77359	531342.0
2	2018-04-06	1	875	INR	5321	158203.0
3	2018-04-11	1	583	INR	21343	187912.0
4	2018-06-18	6	7176	INR	75378	543530.0


```
[10]: df4 = pd.read_excel(r"C:\Users\ASUS\Downloads\restaurant.xlsx")
```

```
[11]: df4.head()
```

```
[11]:
```

	id	name	Country	city	rating	rating_count	cuisine	link	address
0	567335	AB FOODS POINT	India	Abohar	--	Too Few Ratings	Beverages,Pizzas	https://www.swiggy.com/restaurants/ab-foods-po...	AB FOODS POINT, NEAR RISHI NARANG DENTAL CLINI...
1	531342	Janta Sweet House	India	Abohar	4.4	50+ ratings	Sweets,Bakery	https://www.swiggy.com/restaurants/janta-sweet...	Janta Sweet House, Bazar No.9, Circullar Road,...
2	158203	theka coffee desi	India	Abohar	3.8	100+ ratings	Beverages	https://www.swiggy.com/restaurants/theka-coffe...	theka coffee desi, sahtiya sadan road city
3	187912	Singh Hut	India	Abohar	3.7	20+ ratings	Fast Food,Indian	https://www.swiggy.com/restaurants/singh-hut-n...	Singh Hut, CIRCULAR ROAD NEAR NEHRU PARK ABOHAR
4	543530	GRILL MASTERS	India	Abohar	--	Too Few Ratings	Italian-American,Fast Food	https://www.swiggy.com/restaurants/grill-maste...	GRILL MASTERS, ADA Heights, Abohar - Hanumanga...

[12]: df.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 100000 entries, 0 to 99999
Data columns (total 6 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   user_id         100000 non-null  int64
1   name            100000 non-null  object
2   Age             100000 non-null  int64
3   Gender          100000 non-null  object
4   Marital Status  100000 non-null  object
5   Occupation      100000 non-null  object
dtypes: int64(2), object(4)
memory usage: 4.6+ MB
```

[13]: df1.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 371539 entries, 0 to 371538
Data columns (total 3 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   f_id            371531 non-null  object
1   item            371531 non-null  object
2   veg_or_non_veg  371531 non-null  object
dtypes: object(3)
memory usage: 8.5+ MB
```

[14]: df2.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1048575 entries, 0 to 1048574
Data columns (total 5 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   menu_id         1048575 non-null  object
1   r_id            1048575 non-null  int64
2   f_id            1048575 non-null  object
3   cuisine         1048575 non-null  object
4   price           1048574 non-null  float64
dtypes: float64(1), int64(1), object(3)
memory usage: 40.0+ MB
```

[15]: df3.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150281 entries, 0 to 150280
Data columns (total 6 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   order_date      150281 non-null  datetime64[ns]
1   sales_qty       150281 non-null  int64
2   sales_amount    150281 non-null  int64
3   currency        150281 non-null  object
4   user_id         150281 non-null  int64
5   r_id            148664 non-null  float64
dtypes: datetime64[ns](1), float64(1), int64(3), object(1)
memory usage: 6.9+ MB
```

df4.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 148540 entries, 0 to 148539
Data columns (total 9 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   id              148540 non-null  int64
1   name            148454 non-null  object
2   Country         148540 non-null  object
3   city            148540 non-null  object
4   rating          148454 non-null  object
5   rating_count    148454 non-null  object
6   cuisine         148441 non-null  object
7   link            148540 non-null  object
8   address         148454 non-null  object
dtypes: int64(1), object(8)
memory usage: 10.2+ MB
```




```
[22]: df.isnull().sum()
```

```
[22]: user_id      0
      name        0
      Age         0
      Gender      0
      Marital Status  0
      Occupation  0
      dtype: int64
```

```
[23]: df1.isnull().sum()
```

```
[23]: f_id      8
      item      8
      veg_or_non_veg  8
      dtype: int64
```

```
[24]: df2.isnull().sum()
```

```
[24]: menu_id      0
      r_id        0
      f_id        0
      cuisine     0
      price       1
      dtype: int64
```

```
[25]: df3.isnull().sum()
```

```
[25]: order_date      0
      sales_qty      0
      sales_amount   0
      currency       0
      user_id        0
      r_id          1617
      dtype: int64
```

```
[26]: df4.isnull().sum()
```

```
[26]: id      0
      name    86
      Country  0
      city    0
      rating  86
      rating_count  86
      cuisine  99
      link     0
      address  86
      dtype: int64
```



IMPORTING SQL

```
[27]: import sqlite3
import warnings
warnings.filterwarnings('ignore')
```

```
[28]: conn = sqlite3.connect("sqlite.db")
```

```
[29]: df.to_sql("users", conn, if_exists="replace", index=False)
df1.to_sql("encoded-food", conn, if_exists="replace", index=False)
df2.to_sql("encoded-menu (1)", conn, if_exists="replace", index=False)
df3.to_sql("orders", conn, if_exists="replace", index=False)
df4.to_sql("restaurant", conn, if_exists="replace", index=False)
```

```
[29]: 148540
```



IMPORTING SQL

```
[30]: cursor = conn.cursor()
```

```
[31]: cursor.execute ("""select name from sqlite_master where type = 'table';""")
print('list of tables present in database')
table_list = [table[0] for table in cursor.fetchall()]
table_list
```

list of tables present in database

```
[31]: ['users.xlsx',
'encoded-food.csv',
'encoded-menu (1).csv',
'orders.xlsx',
'restaurant.xlsx',
'encoded_food',
'menu',
'users',
'encoded-food',
'encoded-menu (1)',
'orders',
'restaurant']
```



1. How many total orders were placed?

```
[38]: orders = pd.read_sql_query("SELECT COUNT(*) AS total_orders FROM orders",conn)
orders.head()
```

```
[38]:
```

	total_orders
0	150281

2. What is the total revenue across all orders?

```
[39]: orders = pd.read_sql_query("SELECT SUM(sales_amount) AS total_revenue FROM ORDERS WHERE sales_amount > 0",conn)
orders.head()
```

```
[39]:
```

	total_revenue
0	986565018



3. What is the average revenue per order?

```
[40]: orders = pd.read_sql_query("SELECT ROUND(SUM(sales_amount) * 1.0 / COUNT(*),2) AS avg_order_value FROM ORDERS WHERE sales_amount > 0" , conn)
orders.head()
```

```
[40]:
```

	avg_order_value
0	6635.94

4. Which cities have the highest order counts?



```
[41]: orders = pd.read_sql_query(
    "SELECT restaurant.city, COUNT(orders.r_id) AS total_orders \
    FROM orders JOIN restaurant ON orders.r_id = restaurant.id \
    GROUP BY restaurant.city \
    ORDER BY total_orders DESC LIMIT 10",
    conn
)
orders.head()
```

```
[41]:
```

	city	total_orders
0	Bikaner	1666
1	Noida-1	1428
2	Indirapuram,Delhi	1279
3	BTM,Bangalore	1161
4	Rohini,Delhi	1136

5. Which cuisines contribute most revenue?

```
[43]: revenue = pd.read_sql_query("""
        SELECT restaurant.cuisine,
               SUM(orders.sales_amount) AS total_revenue
        FROM orders
        JOIN restaurant
         ON orders.r_id = restaurant.id
        GROUP BY restaurant.cuisine
        ORDER BY total_revenue DESC
        LIMIT 5;
        """, conn)

revenue.head()
```

```
[43]:
```

	cuisine	total_revenue
0	North Indian,Chinese	44956268
1	Indian	42626090
2	North Indian	33670996
3	Chinese	27259296
4	Indian,Chinese	25738592

6. Which cuisines do males vs females prefer?

```
[44]: gender_preference = pd.read_sql_query("""
      SELECT
        users.Gender,
        restaurant.cuisine,
        COUNT(orders.r_id) AS total_orders
      FROM orders
      JOIN users
        ON orders.user_id = users.user_id
      JOIN restaurant
        ON orders.r_id = restaurant.id
      GROUP BY users.Gender, restaurant.cuisine
      ORDER BY total_orders DESC
      """, conn)

gender_preference.head(10)
```

```
[44]:
```

	Gender	cuisine	total_orders
0	Male	North Indian,Chinese	3676
1	Male	Indian	3658
2	Male	Chinese	2881
3	Female	North Indian,Chinese	2799
4	Female	Indian	2759
5	Male	North Indian	2755
6	Male	Indian,Chinese	2530
7	Female	Chinese	2178
8	Female	North Indian	2024
9	Male	South Indian	1870

7. What percentage of orders are veg vs non-veg?

```
[46]: cursor.execute('ALTER TABLE "encoded_food" RENAME TO food')
      conn.commit()
```

```
[48]: veg_or_non_veg = pd.read_sql_query("""
      SELECT food.veg_or_non_veg,
             COUNT(orders.r_id) AS total_orders
      FROM orders
      JOIN menu
        ON orders.r_id = menu.r_id
      JOIN food
        ON food.f_id = menu.f_id
      GROUP BY food.veg_or_non_veg;
      """, conn)

      veg_or_non_veg.head()
```

```
[48]:
```

	veg_or_non_veg	total_orders
0	Non-veg	321885
1	Veg	740856



8. Which age groups spend the most?

```
[50]: age = pd.read_sql_query("SELECT users.age , ROUND(SUM(orders.sales_amount) * 1.0 / COUNT(DISTINCT users.user_id ),2) AS average_spend FROM orders \
JOIN users ON orders.user_id = users.user_id \
GROUP BY users.age \
ORDER BY users.age" , conn)
age.head()
```

```
[50]:
```

	Age	average_spend
0	18	8705.59
1	19	11053.47
2	20	11229.71
3	21	13561.28
4	22	13011.61

9. Which restaurants generate the most revenue? (Top 10 Restaurants)

```
[52]: restaurant = pd.read_sql_query("SELECT restaurant.name , SUM(orders.sales_amount) AS total_revenue FROM orders \
JOIN restaurant \
ON orders.r_id = restaurant.id \
GROUP BY restaurant.name \
ORDER BY total_revenue DESC \
LIMIT 10" , conn)
restaurant.head(10)
```

```
[52]:
```

	name	total_revenue
0	Domino's Pizza	5025266
1	Kouzina Kafe - The Food Court	1958408
2	Sweet Truth - Cake and Desserts	1952881
3	Pizza Hut	1787100
4	Huber & Holly	1668292
5	Biryani House	1655736
6	Baskin Robbins	1627731
7	KFC	1605569
8	McCafe by McDonald's	1541849
9	Janta Snacks	1510944



10. How does order volume change with price – which items are overpriced or underpriced?

```
[55]: price = pd.read_sql_query("""
        SELECT menu.cuisine,
               menu.price,
               COUNT(orders.r_id) AS total_orders,
               ROUND(SUM(orders.sales_amount), 2) AS total_revenue
        FROM orders
        JOIN menu
           ON orders.r_id = menu.r_id
        GROUP BY menu.cuisine, menu.price
        HAVING COUNT(orders.r_id) > 50
        ORDER BY menu.price DESC
        """, conn)

price.head()
```

	cuisine	price	total_orders	total_revenue
0	Bakery	2800.0	77	121741.0
1	Bakery	2750.0	63	100759.0
2	North Indian,Barbecue	1079.0	74	471733.0
3	Bakery	1000.0	88	246873.0
4	Desserts,Bakery	999.0	57	878800.0



Months

January

February

March



Active Customers

100K

Total Orders

150K

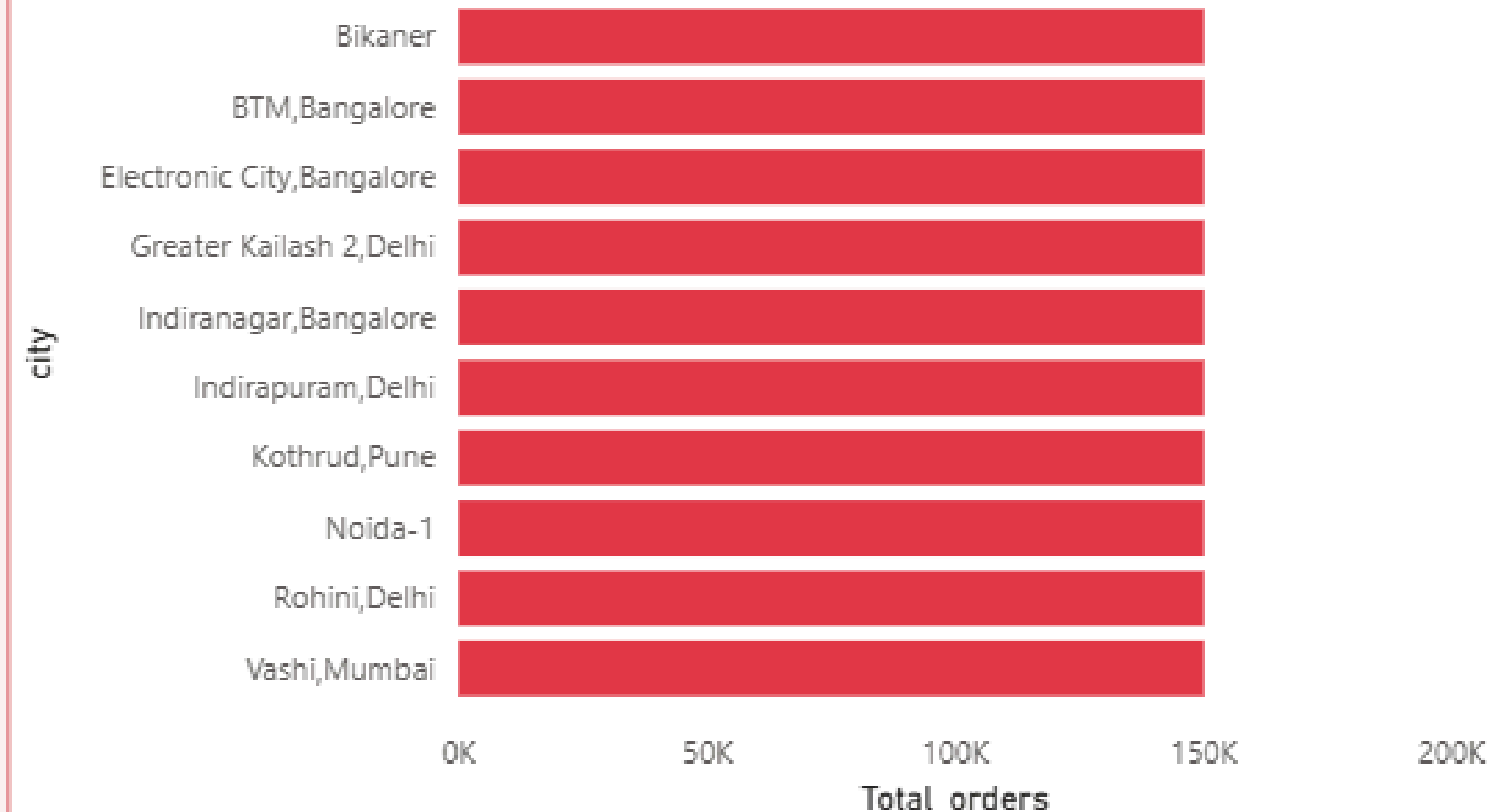
Average Order Value

1.37K

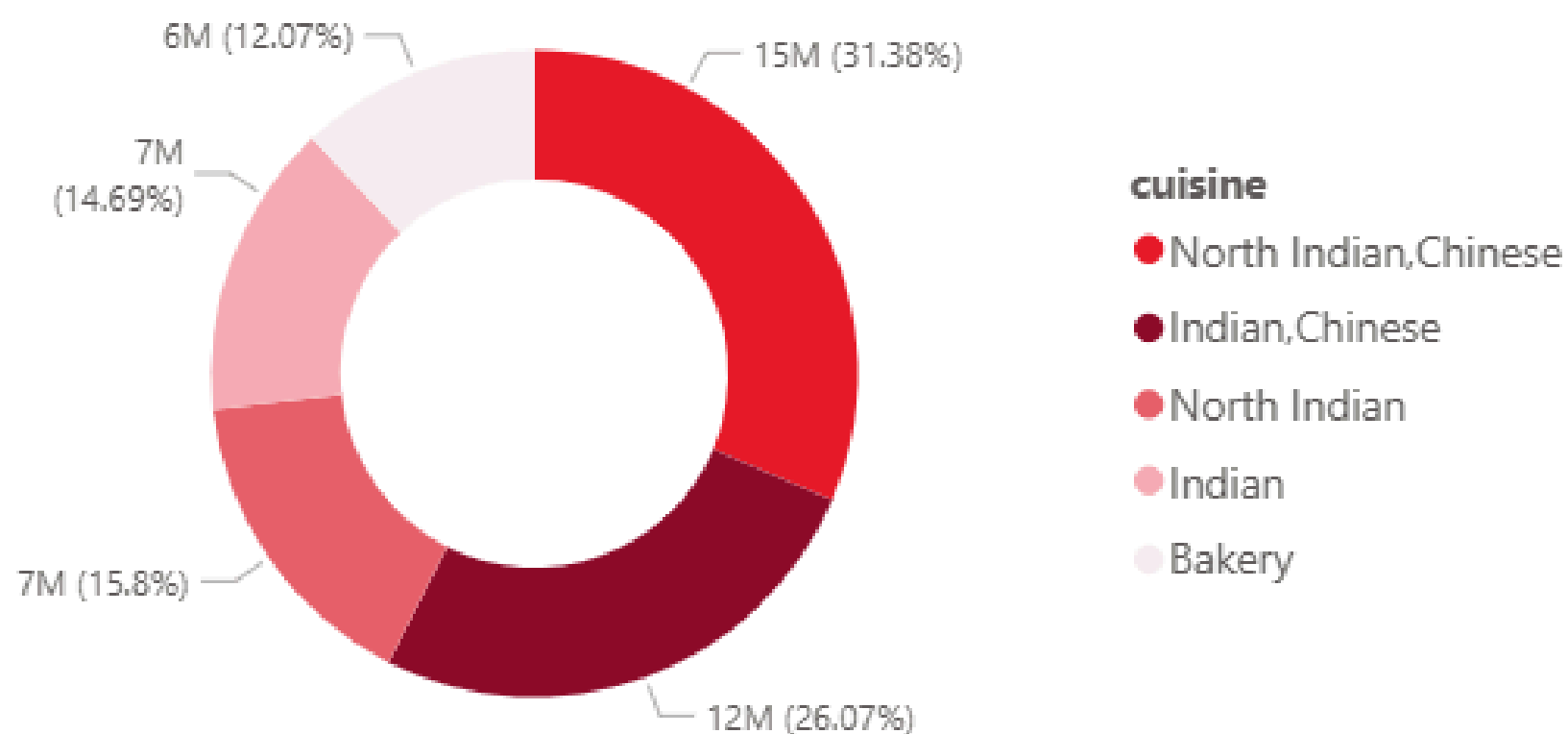
Total Revenue

206M

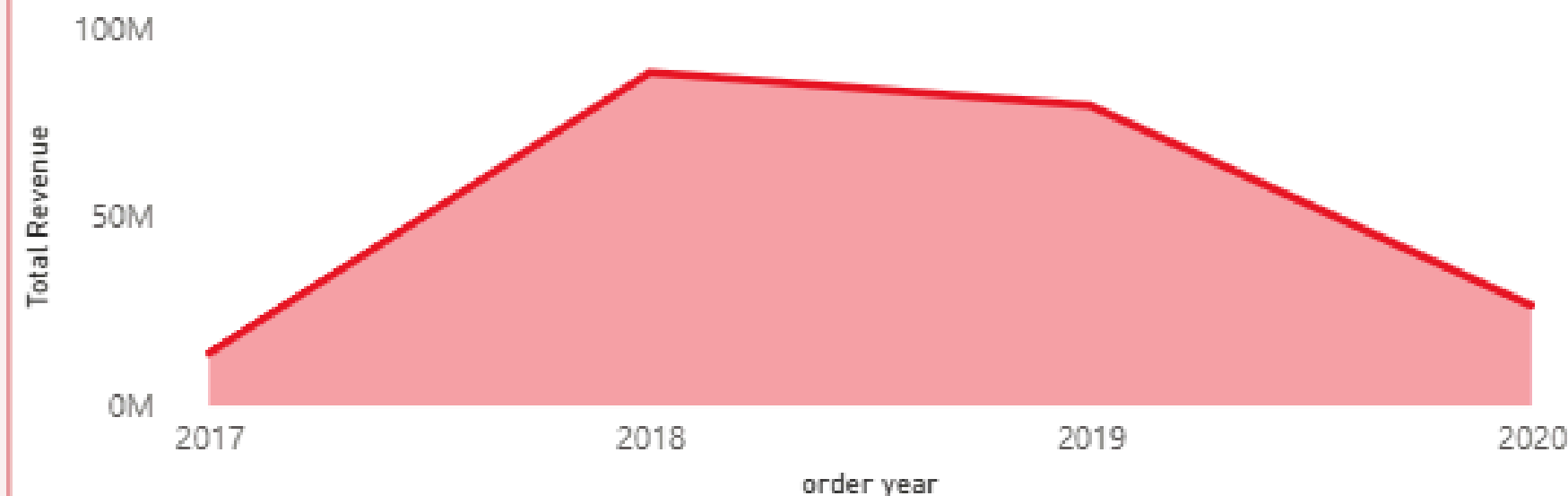
Top 10 Clty By Orders



Top Cuisine



Revenue Trend By Years





Chinese

Indian

Indian,Chinese



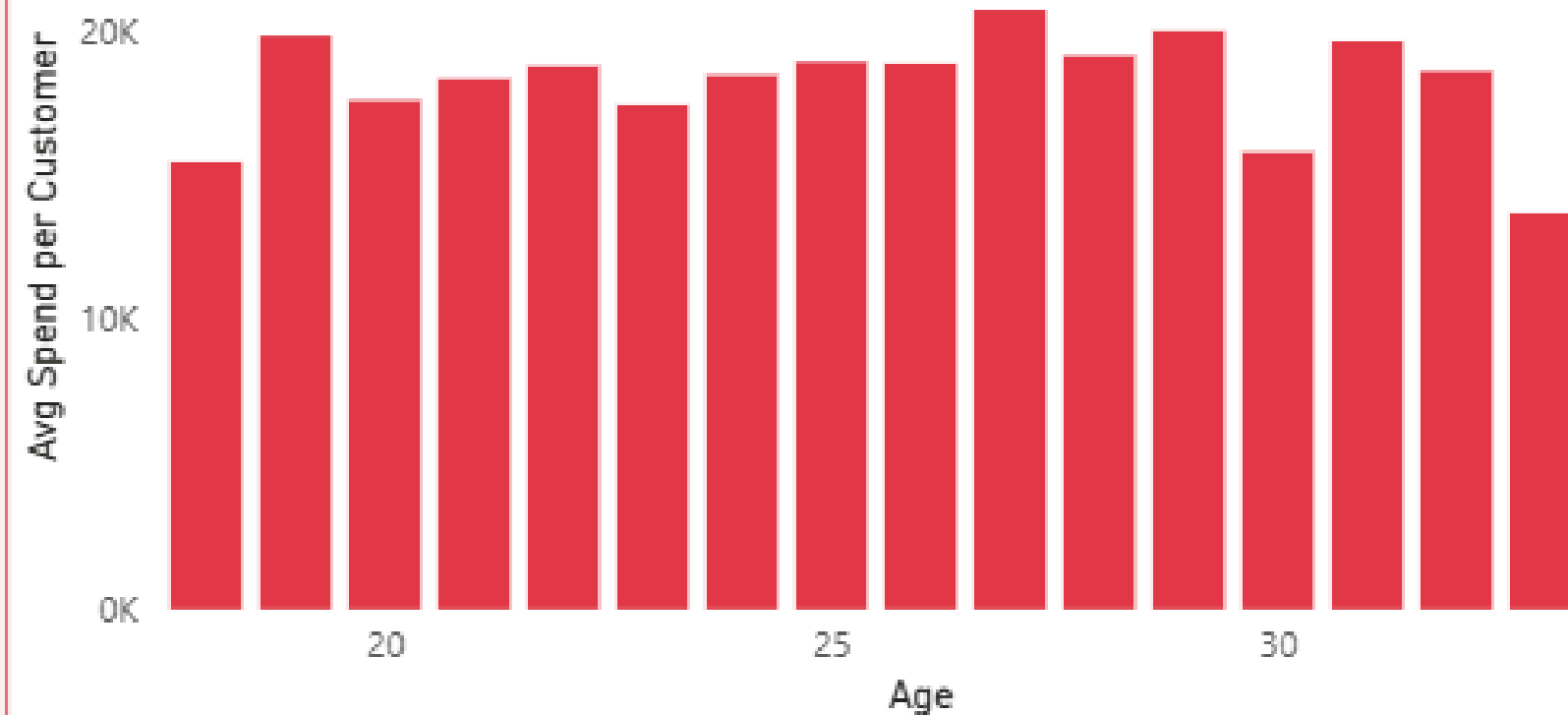
Average Spend By Customer

18.53K

Total Customer

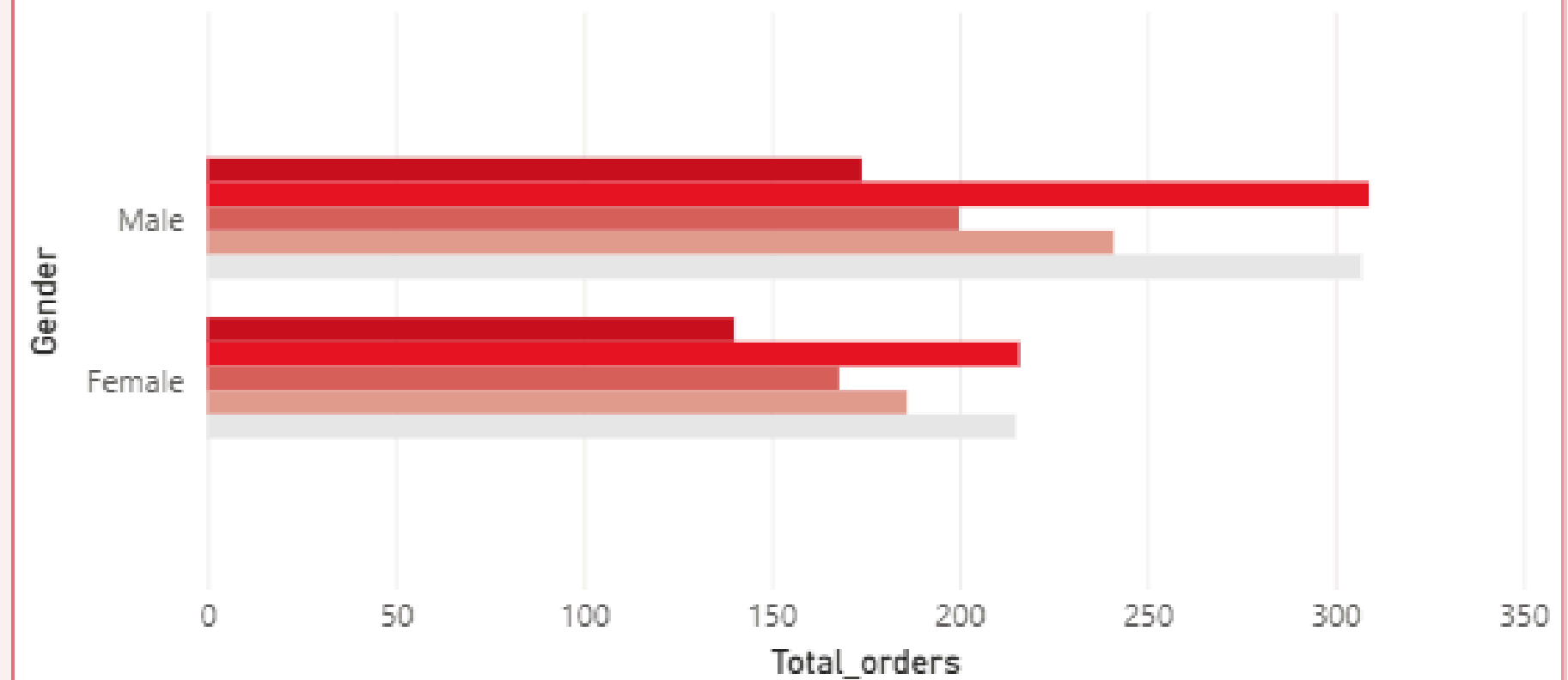
71.25K

Avg Spend per Customer by Age

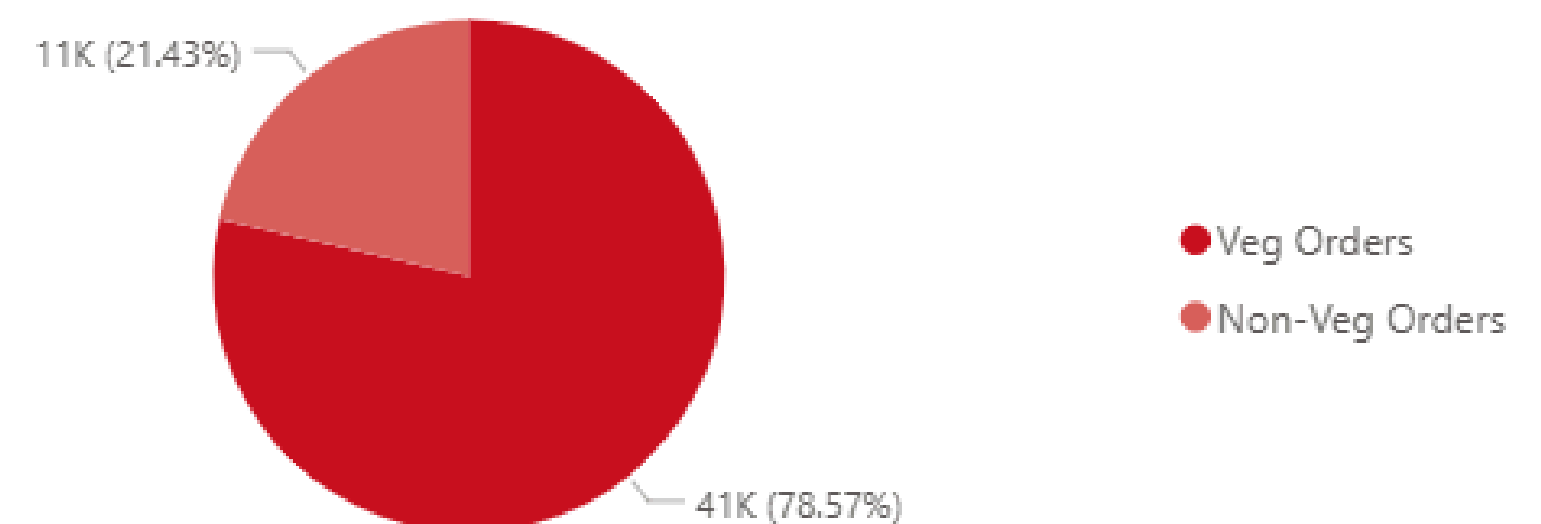


Gender Wise cuisine Preference

Cuisine ● Chinese ● Indian ● Indian,Chinese ● North Indian ● North Indian,Chinese



Veg Orders Vs Non-Veg Orders





Average Price Per Item

196.91

Total Items In Menu

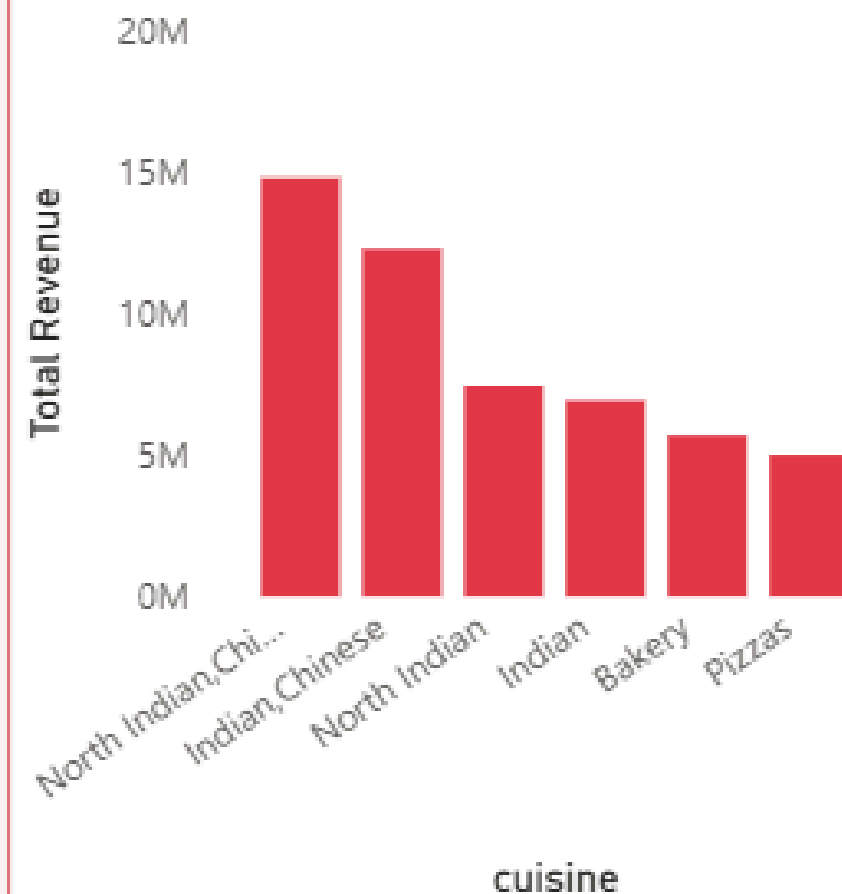
1,436

Best Restaurant Name

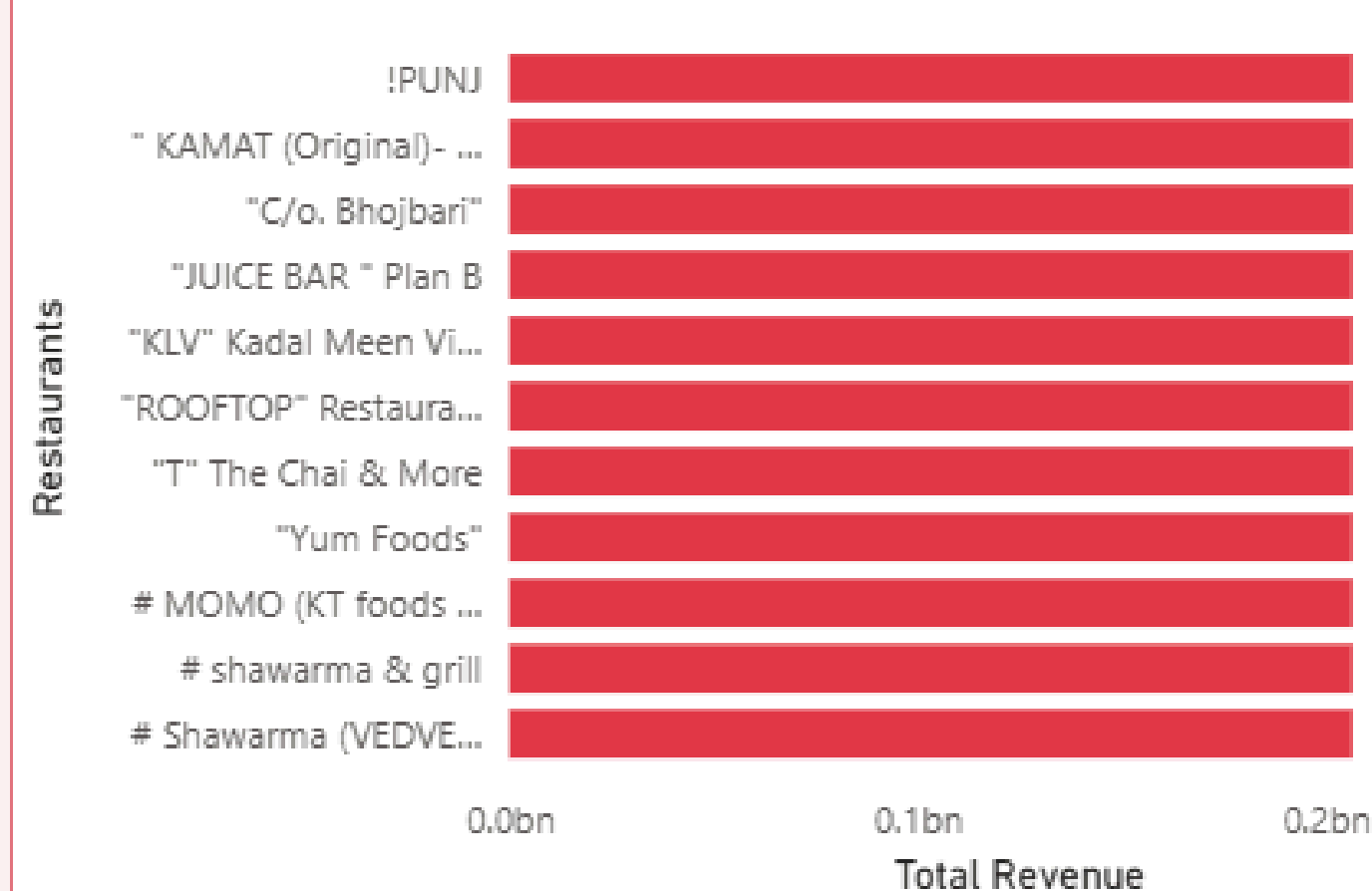
Agrawal Sweets

Restaurants	rating	rating_count	city
!PUNJ	--	Too Few Ratings	Mayur Vihar, Delhi
" KAMAT (Original)- Siripuram"	4	50+ ratings	Dwarka Nagar, Vizag
"C/o. Bhojbari"	--	Too Few Ratings	Siliguri
"JUICE BAR " Plan B	--	Too Few Ratings	Erode
"KLV" Kadal Meen Virpanai Nilayam	--	Too Few Ratings	Dindigul
"ROOFTOP" Restaurant By Hotel Silver suits	3.6	20+ ratings	Bardhaman
"T" The Chai & More	4.7	100+ ratings	Navrangpura, Ahmedabad
"Yum Foods"	--	Too Few Ratings	Alambagh, Lucknow
# MOMO (KT foods & Services)	--	Too Few Ratings	Jodhpur
# shawarma & grill	--	Too Few Ratings	Perambur, Chennai

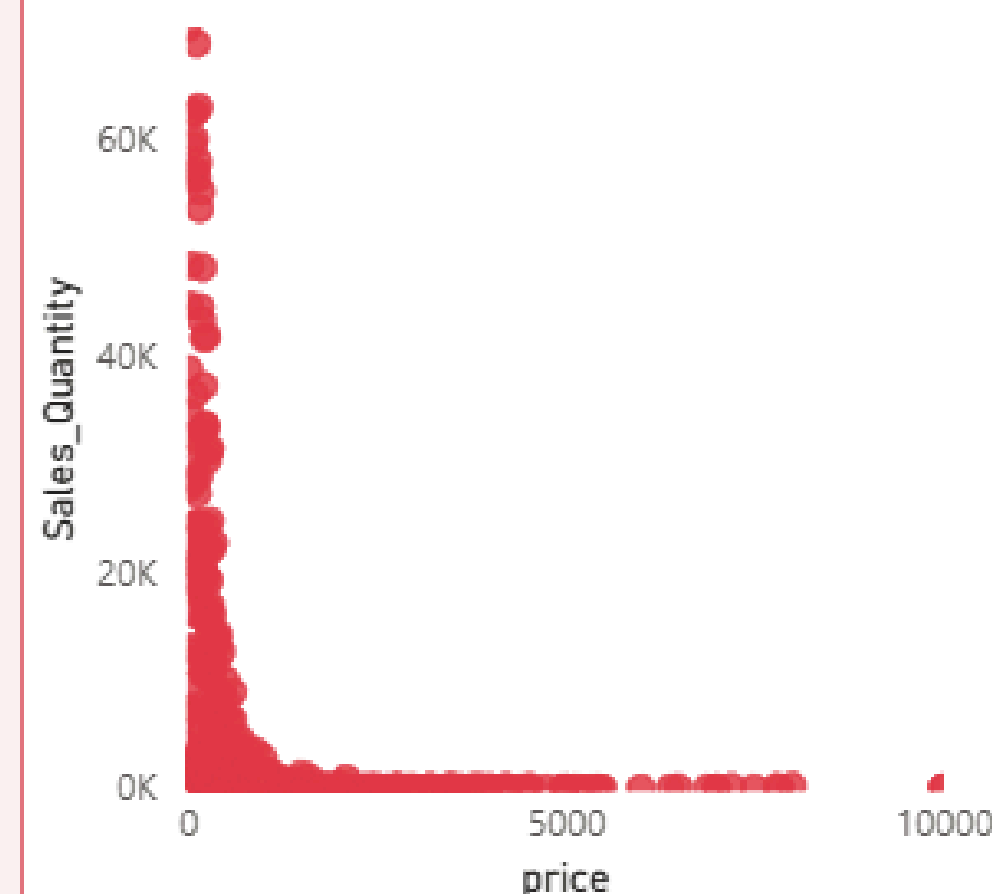
Top 6 cuisines By Revenue



Top Restaurants By Revenue



Price vs Quantity Sold





ANALYSIS REPORT

zomato

Zomato's data shows how customer demand, restaurant performance, and regional growth shape the food delivery market. With 150K orders, 100K active customers, and revenue crossing 206M, the platform has built strong traction across cities like Bangalore, Delhi, Pune, and Mumbai. Popular cuisines such as North Indian-Chinese (31%), Indian-Chinese (26%), and North Indian (16%) emerge as the biggest revenue drivers, reflecting customer preference for comfort foods. Revenue trends also highlight a peak in 2018–2019, signaling a phase of strong growth.

Customer behavior adds another layer of insight. The average spend per customer is 18.5K, with people in their 20s and early 30s spending the most. Gender patterns reveal that males order more Chinese and Indian food, while females lean toward North Indian cuisines. Food choices are also clear—non-vegetarian dishes account for 78% of orders, while vegetarian food makes up 21%, pointing to a strong tilt in demand.

On the restaurant side, Zomato hosts 1,436 menu items, with an average price of 196.91 per dish. Outlets like Agrawal Sweets lead in revenue, and the top cuisines contribute the bulk of sales. Pricing trends reveal customer sensitivity to high menu rates, while city-level analysis highlights Bikaner, Bangalore, and Delhi as hotspots. This project helped me use SQL and Power BI to turn raw transactional data into meaningful insights, uncovering profit drivers, customer patterns, and growth opportunities for the food delivery industry.



**ZOMATO
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SHUBHAM SHARMA

THANK YOU

FOR YOUR TIME AND ATTENTION



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